

1 (i)	$y_n = x_n - 1 = \frac{x_{n-1}x_{n-2} + 1}{x_{n-1} + x_{n-2}} - 1$ $= \frac{x_{n-1}x_{n-2} + 1 - x_{n-1} - x_{n-2}}{x_{n-1} + x_{n-2}}$ $= \frac{(x_{n-1} - 1)(x_{n-2} - 1)}{x_{n-1} + x_{n-2}}$ $= \frac{y_{n-1}y_{n-2}}{y_{n-1} + y_{n-2} + 2} \text{ for } n \geq 3.$
(ii)	Let $P(n)$ be the statement $0 < y_n < 1$ for $n \geq 1$. Clearly, $P(1)$ and $P(2)$ are true since $0 < y_1 = y_2 = \frac{1}{2} < 1$. Assume that $P(1), \dots, P(k-1)$ are true for some $k \in \mathbb{Z}^+, k \geq 3$. $y_k = \frac{y_{k-1}y_{k-2}}{y_{k-1} + y_{k-2} + 2} \text{ and since all the terms are positive on the RHS (by induction hypothesis), we have also } y_k > 0. \text{ Also,}$ $y_k = \frac{y_{k-1}y_{k-2}}{y_{k-1} + y_{k-2} + 2} < \frac{1(1)}{0+0+2} = \frac{1}{2} < 1$ So $P(k)$ is true. Since $P(1)$ and $P(2)$ are true, by mathematical induction, $P(n)$ is true for all $n \geq 1$.
(iii)	$y_n = \frac{y_{n-1}y_{n-2}}{y_{n-1} + y_{n-2} + 2}$ $< y_{n-1} \frac{1}{0+0+2} = \frac{1}{2} y_{n-1}$ since (ii) gives $0 < y_n < 1$ for $n \geq 1$.
(iv)	We have $y_1 + y_2 + y_3 + \dots + y_n < y_1 + y_2 + \frac{1}{2}y_2 + \frac{1}{4}y_2 + \dots + \frac{1}{2^{n-2}}y_2$ $= \frac{1}{2} + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots + \frac{1}{2^{n-1}} < \frac{1}{2} + \frac{\frac{1}{2}}{1 - \frac{1}{2}} = \frac{3}{2}.$ Since all terms in the series are positive, and the sequence of partial sums is bounded, the series converge. OR $y_1 + y_2 + y_3 + \dots + y_n < \frac{1}{2} + \lim_{N \rightarrow \infty} \sum_{n=1}^N \left(\frac{1}{2}\right)^n$ and since the limit on the RHS exist as it is a sum of GP with ratio $0.5 < 1$, by comparison test, the series on the LHS converges.
(v)	(1) holds since $1 < x_n$ for all n (2) holds by AM-GM inequality, and the inequality is strict as not all terms are equal for $n \geq 3$. (3) We have from (iv) that $\frac{x_1 + x_2 + \dots + x_n}{n} = \frac{y_1 + \dots + y_n + n}{n} < \frac{n + \frac{3}{2}}{n} = 1 + \frac{3}{2n}.$

2 (i)	$0 \leq \frac{x^6}{(x^2+1)^4} \leq \frac{x^6}{(2\sqrt{x^2})^4} \text{ (AM-GM)}$ $= \frac{x^2}{16} \leq \frac{1}{16} \text{ for } x \leq 1$
(ii)	$\frac{d}{dx} \left(\frac{Ax^5 + Bx^3 + Cx}{(x^2+1)^3} \right) = \frac{(x^2+1)^3 (5Ax^4 + 3Bx^2 + C) - (Ax^5 + Bx^3 + Cx) 3(x^2+1)^2 (2x)}{(x^2+1)^6}$ $= \frac{(5A-6A)x^6 + (5A+3B-6B)x^4 + (3B+C-6C)x^2 + C}{(x^2+1)^4}$ $= \frac{-Ax^6 + (5A-3B)x^4 + (3B-5C)x^2 + C}{(x^2+1)^4}$ $\therefore 1 = (D-A)x^6 + (5A-3B)x^4 + (3B-5C)x^2 + C$ $C=1, 3B=5C, 5A=3B, D=A \Rightarrow B=\frac{5}{3}, A=1=D$
(iii)	$\int_0^1 \frac{1}{(x^2+1)^4} dx = \left[\frac{x^5 + \frac{5}{3}x^3 + x}{(x^2+1)^3} \right]_0^1 + \int_0^1 \frac{x^6}{(x^2+1)^4} dx$ $\left(\frac{x^5 + \frac{5}{3}x^3 + x}{(x^2+1)^3} \right)_0^1 = \frac{1 + \frac{5}{3} + 1}{2^3} = \frac{11}{24} \quad \text{AND} \quad 0 \leq \int_0^1 \frac{x^6}{(x^2+1)^4} dx \leq \int_0^1 \frac{1}{16} dx = \frac{1}{16}$ $\therefore \frac{11}{24} \leq \int_0^1 \frac{1}{(x^2+1)^4} dx \leq \frac{11}{24} + \frac{1}{16} = \frac{25}{48}$ But $\frac{1}{(x^2+1)^4}$ is even, so $\int_{-1}^1 \frac{1}{(x^2+1)^4} dx = 2 \int_0^1 \frac{1}{(x^2+1)^4} dx$. $\therefore \frac{11}{12} = 2 \left(\frac{11}{24} \right) \leq \int_{-1}^1 \frac{1}{(x^2+1)^4} dx \leq 2 \left(\frac{25}{48} \right) = \frac{25}{24}$.

3(i)	Since $\frac{d}{dx} f'(x) = f''(x) > 0$, f' is an increasing function. Hence it is one to one.
(ii)	Let the tangent with gradient p be tangent to the curve $y = f(x)$ at $(x_p, f(x_p))$. Then $f'(x_p) = p$, so $x_p = (f')^{-1}(p)$. This tangent has equation $y - f(x_p) = p(x - x_p) \Rightarrow y = px - (px_p - f(x_p)).$ Therefore $g(p) = px_p - f(x_p)$.
(iii)	For a fixed p, let $h_p(x) = px - f(x)$. We have $h_p'(x) = p - f'(x) \quad \text{and} \quad h_p''(x) = -f''(x).$ For stationary points, $h_p'(x) = 0$, so $x = (f')^{-1}(p) = x_p$ as in (ii). This gives the stationary value $h_p(x_p) = x_p p - f(x_p) = g(p)$. Since $h_p''(x_p) = -f''(x_p) < 0$, this is a maximum value via the second derivative test.

Note	An important property of the Legendre transform is that g is also a convex function. There was originally a part of the question to show this, but we decided to drop it. The proof is reproduced here for your interest. To show that g is convex, we need to check that $g''(p) > 0$ for all p. Differentiating $g(p) = px_p - f(x_p)$, and remembering that x_p is a function of p, we get $\begin{aligned} g'(p) &= \frac{d}{dp}(px_p - f(x_p)) \\ &= x_p + p \frac{dx_p}{dp} - \frac{d}{dx_p} f(x_p) \frac{dx_p}{dp} \quad (\text{product and chain rules}) \\ &= x_p + (p - f'(x_p)) \frac{dx_p}{dp} \\ &= x_p. \end{aligned}$ Differentiating the relation $p = f'(x_p)$ with respect to p gives $\begin{aligned} 1 &= \frac{d}{dp} f'(x_p) \\ &= \frac{d}{dx_p} (f'(x_p)) \frac{dx_p}{dp} \quad (\text{chain rule}) \\ &= f''(x_p) \frac{dx_p}{dp}. \end{aligned}$ Therefore $g''(p) = \frac{dx_p}{dp} = \frac{1}{f''(x_p)} > 0$.
(iv)	We have $f'(x) = x^{m-1}$ and $f''(x) = (m-1)x^{m-2} > 0$ for all $x > 0$, so f is convex. The domain of g is the range of f', namely $(0, \infty)$. For $p \in (0, \infty)$, $f'(x_p) = p \Rightarrow x_p^{m-1} = p \Rightarrow x_p = p^{\frac{1}{m-1}}.$ Therefore $\begin{aligned} g(p) &= px_p - f(x_p) \\ &= pp^{\frac{1}{m-1}} - \frac{1}{m} p^{\frac{m}{m-1}} \\ &= \left(1 - \frac{1}{m}\right) p^{\frac{m}{m-1}} \\ &= \frac{p^n}{n}. \end{aligned}$
(v)	Since $g(p)$ is the maximum value of $xp - f(x)$, $g(p) \geq xp - f(x) \Rightarrow xp \leq f(x) + g(p)$ for all $x > 0$ and $p > 0$ (cf the domain of g in (v)). Using the result of (v) gives the desired inequality $xp \leq \frac{x^m}{m} + \frac{p^n}{n}$. For equality, $x = x_p = p^{\frac{1}{m-1}}$, which simplifies to $x^m = p^{\frac{m}{m-1}} = p^n$.

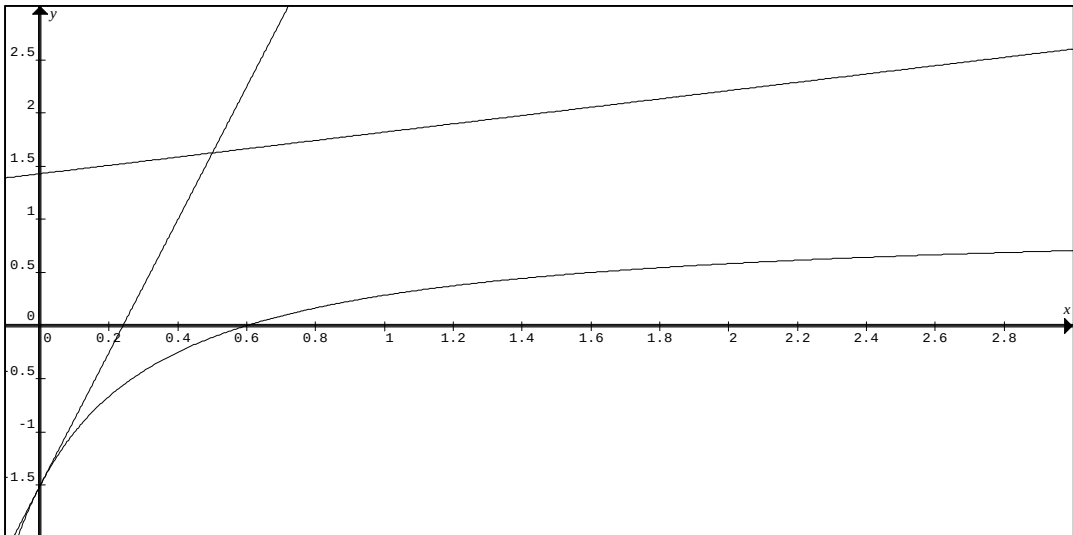
4 (i)	Consider a set with $(n+1)$ elements, say $a_1, a_2, \dots, a_n, a_{n+1}$. Then the RHS is the number of possible subsets of size 3. The LHS shows that there are $n-1$ cases to choose these subsets. Case 1: a_1 is either included in subsets of size 3 or not included. If it is included there are $\binom{n}{2}$ ways to choose the other 2 members of a subset. Case 2: If a_1 is not included we consider a_2. If a_2 is included (but not a_1), there are $\binom{n-1}{2}$ ways to choose the other 2 members of a subset. We continue until Case $n-1$, where $a_1, a_2, \dots, a_{n-3}, a_{n-2}$ are not included but a_{n-1} is. Then there is $\binom{2}{2} = 1$ way to choose the other two members which are a_n and a_{n+1}. By AP, the LHS is $\sum_{k=2}^n \binom{k}{2}$.
(ii)(a)	If there are n digits summing to 9, it is the same as having 9 identical balls distributed to n distinct boxes but the first digit must be non-zero. So we place one ball into the first box to ensure that the first digit is non-zero, and distribute 8 identical balls to n distinct boxes. If each ball is treated as a '0' and each vertical bar separating adjacent boxes as a '1', then each way of distribution becomes a $n+8-1$-digit binary sequence with $n-1$ '1's. By BP, the number of n digit good integers is $\binom{n+8-1}{n-1} = \binom{n+7}{n-1} = \binom{n+7}{8}$.
(ii)(b)	We can see that $\binom{1+7}{8} + \binom{2+7}{8} + \dots + \binom{6+7}{8}$ will give the total number of 6-digit good integers. By an identity similar to (i), we have $\sum_{k=8}^{13} \binom{k}{8} = \binom{14}{9} = 2002$. Hence $m = 2003$.
(ii)(c)	Since a_{2014} is clearly a 7-digit good integer, we see that we are looking for the 12th smallest 7-digit good integer ($2014-2002=12$). Since the first digit must be 1, and there are $\binom{7}{7} + \binom{8}{7} = 9$ one or two-digit numbers summing to 8, we can see that $a_{2012} = 1000107, a_{2013} = 1000116, a_{2014} = 1000125$.
(iii)(a)	Let A be a particular object among the r distinct objects. Then there are two cases, either A is alone in one box, or A is not alone in one box. Case 1: there remain $r-1$ objects to be distributed into 2 identical boxes such that no box is empty. The number of ways to distribute $r-1$ objects into 2 distinct boxes is 2^{r-1}. Since there are two ways for the $r-1$ objects to be all in one box such that the other is empty, there are $2^{r-1} - 2$ ways. Clearly, since the boxes are identical and not distinct, the number of ways is $\frac{1}{2}(2^{r-1} - 2) = 2^{r-2} - 1$.

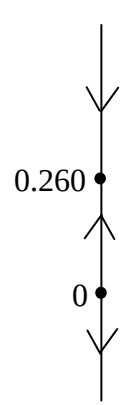
	Case 2: If A is not alone in the box, then we first place $r-1$ objects to be distributed into 3 identical boxes such that no box is empty where there are $S(r-1)$ ways. Since there are 3 ways for A to be placed in the boxes, by MP we have $3S(r-1)$. By AP, $S(r) = 2^{r-2} - 1 + 3S(r-1)$.
(iii)(b)	There are 9 two-digit good integers. We want to place 9 distinct objects into 3 identical and non-empty sets. Using the recurrence, $\sum_{r=2}^7 3^{7-r} (2^r) - \sum_{r=0}^5 3^r + 3^6 = 3025.$ Alternatively, the formula $\frac{1}{2}(3^8 + 1) - 2^8 = 3025$ may be used.

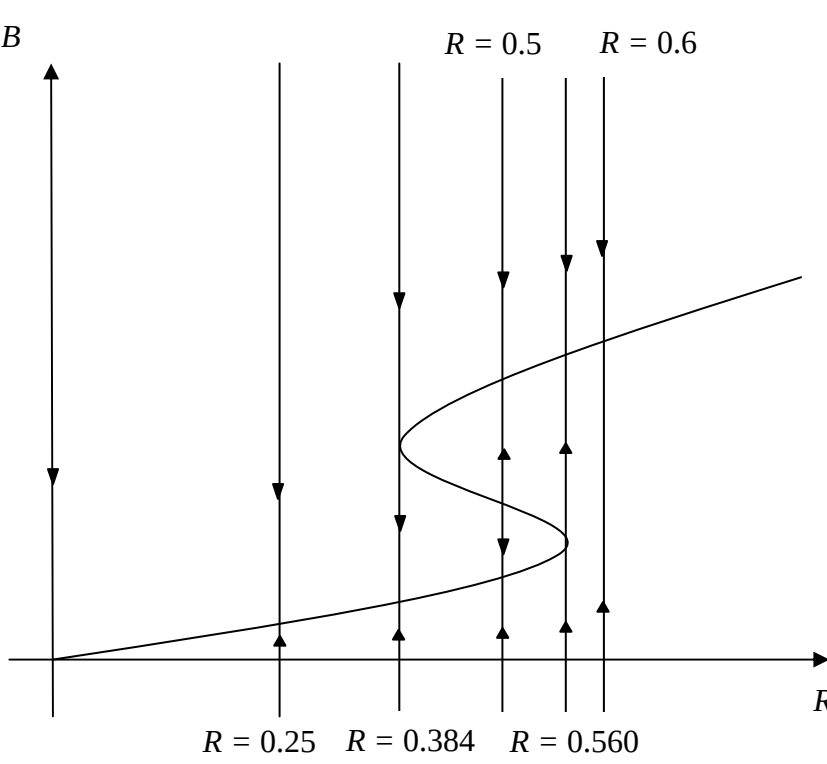
5 (i)	Clearly, $G(1) = 2$ as $\{1, 2\}$ and $\{2, 1\}$ are both great. A check on the 24 permutations of $(1, 2, 3, 4)$ shows that amongst <u>(1, 2, 3, 4)</u> (1, 2, 4, 3) (1, 3, 2, 4) (1, 3, 4, 2) (1, 4, 2, 3) <u>(1, 4, 3, 2)</u> (2, 1, 3, 4) <u>(2, 1, 4, 3)</u> (2, 3, 1, 4) <u>(2, 3, 4, 1)</u> (2, 4, 1, 3) (2, 4, 3, 1) (3, 1, 2, 4) (3, 1, 4, 2) <u>(3, 2, 1, 4)</u> (3, 2, 4, 1) <u>(3, 4, 1, 2)</u> (3, 4, 2, 1) <u>(4, 1, 2, 3)</u> (4, 1, 3, 2) (4, 2, 1, 3) (4, 2, 3, 1) (4, 3, 1, 2) <u>(4, 3, 2, 1)</u> only those underlined are not great. Hence $G(2) = 16$.
(ii)	Since k and $k+n$ are neighbours, consider them as a group. There will be $2n-1$ groups, which have $(2n-1)!$ ways to arrange amongst themselves. Within the group, k and $k+n$ can be in any position, hence we have $X_k = 2(2n-1)!$
(iii)	Since there must be m groups of 2, we have a total of $2n-m$ groups, which have $(2n-m)!$ ways to arrange amongst themselves. Within the group, there are again 2 ways to arrange the neighbours, and thus $ X_{i_1} \cap X_{i_2} \cap \dots \cap X_{i_m} = 2^m (2n-m)!$
(iv)	By the Principle of Inclusion and Exclusion, $G(n) = \sum_{k=1}^n X_k - \sum_{1 \leq k < m \leq n} X_k \cap X_m + \sum_{1 \leq k < l < m \leq n} X_k \cap X_l \cap X_m - \dots$ However, the terms are strictly decreasing if and only if $2^m (2n-m)! > 2^{m+1} (2n-m-1)! \Leftrightarrow (2n-m) > 2.$ Since $m \leq n$ and $n \geq 2$, letting $a_m = \sum_{1 \leq i_1 < i_2 < \dots < i_m \leq n} X_{i_1} \cap X_{i_2} \cap \dots \cap X_{i_m}$ $G(n) = \sum_{k=1}^n X_k - \sum_{1 \leq k < m \leq n} X_k \cap X_m + (a_3 - a_4) + \dots + (-1)^{n+1} X_1 \cap \dots \cap X_n $ $\geq \sum_{k=1}^n X_k - \sum_{1 \leq k < m \leq n} X_k \cap X_m $ since all the terms in brackets are positive after the first two.

(v)	From (iv), $G(n) \geq \sum_{k=1}^n X_k - \sum_{1 \leq k < m \leq n} X_k \cap X_m $ $= 2n(2n-1)! - \binom{n}{2} 4(2n-2)!$ $= (2n)! - \frac{(n-1)(2n)!}{2n-1} = (2n)! \frac{n}{2n-1} > \frac{(2n)!}{2}.$
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6(i)	Integrating factor $I(x) = e^{\int 2x dx} = e^{x^2}$ $\therefore \frac{d}{dx} (e^{x^2} y) = x e^{x^2}$ $e^{x^2} y = \int x e^{x^2} dx = \frac{1}{2} e^{x^2} + C$ $y = \frac{1}{2} + C e^{-x^2}$
(ii)	$y^2 = z \Rightarrow 2y \frac{dy}{dx} = \frac{dz}{dx}$. Hence $y \frac{dy}{dx} + xy^2 = \frac{x}{2} \Leftrightarrow 2y \frac{dy}{dx} + 2xy^2 = x$ $\Leftrightarrow \frac{dz}{dx} + 2xz = x.$ From (i), $z = y^2 = \frac{1}{2} + C e^{-x^2}$. $y = 0$ when $x = 1$ means $0 = \frac{1}{2} + \frac{C}{e} \Rightarrow C = -\frac{e}{2} \Rightarrow y^2 = \frac{1}{2}(1 - e^{-x^2+1})$.
(iii)	Characteristic equation: $4r^2 - 4r + 1 = 0$ $(2r-1)^2 = 0$ $\Rightarrow r = \frac{1}{2}$ General solution: $y = A e^{x/2} + B x e^{x/2}$. Substituting in the given conditions and from $y' = \frac{A}{2} e^{x/2} + B e^{x/2} + \frac{B}{2} x e^{x/2}$, we have $1 = A$ $-\frac{3}{2} = \frac{1}{2} + B \Rightarrow B = -2.$ $\therefore y = e^{x/2} - 2x e^{x/2}$ When $x = \ln 4$, $y = e^{\ln 4/2} - 2 \ln 4 e^{\ln 4/2} = 2 - 2 \ln 4(2) = 2 - 8 \ln 2$.

7 (i)	$\frac{dy}{dx} = (y-1)^2 \Rightarrow \int \frac{1}{(y-1)^2} dy = \int dx$ $\Rightarrow -\frac{1}{y-1} = x + c$ When $x = 0, y = -1.5 \Rightarrow c = -\frac{1}{-2.5} = \frac{2}{5}$. $\therefore y-1 = -\frac{1}{x + \frac{2}{5}} = -\frac{5}{5x+2}.$ When $x = 1, y = 1 - \frac{5}{7} = \frac{2}{7}$.																				
(ii)	Using the Euler Method, $y(0.5) \approx y(0) + 0.5(y(0)-1)^2 = -1.5 + 0.5(6.25) = 1.625$ $y(1) \approx y(0.5) + 0.5(y(0.5)-1)^2 = 1.625 + 0.5(0.390625) = 1.82$ (to 3 sf)																				
(iii)	As seen in the sketch below, the initial tangent to the curve is steep, which results in the first approximation to be already exceed the asymptotic value of y . 																				
(iv)	The completed table is shown below. <table><tr><th>x</th><th>y</th><th>$(y-1)^2$</th><th>$\tilde{y}$</th><th>$\frac{\Delta y}{\Delta x}$</th></tr><tr><td>0</td><td>-1.5</td><td>6.25</td><td>1.625</td><td>3.3203</td></tr><tr><td>0.5</td><td>0.16015</td><td>0.70535</td><td>0.51282</td><td>0.47135</td></tr><tr><td>1</td><td>0.39582 0.400 (3 sf)</td><td></td><td></td><td></td></tr></table>	x	y	$(y-1)^2$	$\tilde{y}$	$\frac{\Delta y}{\Delta x}$	0	-1.5	6.25	1.625	3.3203	0.5	0.16015	0.70535	0.51282	0.47135	1	0.39582 0.400 (3 sf)			
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8 (i)	Q is the carrying capacity.
(ii)	$\frac{dB}{dt} = RB \left(1 - \frac{B}{Q} \right) - \frac{B^2}{1+B^2}$ $= RB - \frac{R}{Q} B^2 - B^2 (1 - B^2 + B^4 - \dots)$ $\approx RB$ for B sufficiently small, neglecting terms in B^2 and above.
(iii)	Hence for $B > 0$ sufficiently small, $\frac{dB}{dt} > 0$ and for $B < 0$ sufficiently small, $\frac{dB}{dt} < 0$. Thus the extinction equilibrium is unstable.
(iv)	At equilibrium, $\frac{dB}{dt} = RB \left(1 - \frac{B}{10} \right) - \frac{B^2}{1+B^2} = 0$ $\Rightarrow B \left[R \left(1 - \frac{B}{10} \right) - \frac{B}{1+B^2} \right] = 0$ $\Rightarrow B = 0 \quad \text{or} \quad R \left(1 - \frac{B}{10} \right) = \frac{B}{1+B^2}.$
(v)	$\frac{dB}{dt} = B \left[R \left(1 - \frac{B}{10} \right) - \frac{B}{1+B^2} \right]$ For $R = 0.25$, the equilibria are given by $B = 0$ and $B = 0.260$ (3 sf) For $0 < B < 0.260$, $\frac{dB}{dt} = B \left[R \left(1 - \frac{B}{10} \right) - \frac{B}{1+B^2} \right] > 0$ since the line is above the curve. For $B > 0.260$, $\frac{dB}{dt} = B \left[R \left(1 - \frac{B}{10} \right) - \frac{B}{1+B^2} \right] < 0$. Hence the phase diagram is given as follow:  $B = 0$ is unstable and $B = 0.260$ is stable.

(vi)	 The diagram is a bifurcation plot with B on the vertical axis and R on the horizontal axis. A curve starts at the origin and increases. Vertical lines are drawn at $R = 0.25$, $R = 0.384$, $R = 0.560$, $R = 0.5$, and $R = 0.6$. Arrows on these lines indicate the direction of the system's evolution. At $R = 0.384$, a bifurcation occurs where a new stable equilibrium appears. At $R = 0.560$, another bifurcation occurs where the lower two equilibrium points vanish, leaving only a single stable equilibrium at a high value of B.
Note	As the budworm population grows and B increases, the line from the left hand side of the equation becomes tangent to the curve from the right hand side of the equation at $R = 0.384$. This is a bifurcation with a new equilibrium appearing. As R increases more, this results in two new equilibria with the lower one being unstable and the upper one being stable. However, the solution of the differential equation remains trapped near the lower stable equilibrium. Thus, the spruce budworm population remains endemic, and the forest remains healthy and can grow with the bird population keeping the budworms sufficiently in check. As R continues to increase to $R = 0.560$, we see another bifurcation and the lower two equilibrium points vanishing. This results in only having a single stable equilibrium with a high value. At this point there is an outbreak of the spruce budworms in large numbers as they saturate the ability of the predators to control their population and achieve very high numbers. The population goes to a carrying capacity, which causes massive defoliation of the forests. In the model this changes the slow dynamics of the parameter R, and it begins to decrease. As defoliation continues, we again pass through the bifurcation at $R = 0.560$, but now the solution remains at the larger equilibrium, so the health of the forest continues to decline. This continues until the budworms so severely damage the forest that their population collapses. This is shown in the model when the second saddle node bifurcation occurs at $R = 0.384$, and the only remaining equilibrium is the stable endemic equilibrium. From this point, we see that this cycle between the growth of the forest and the outbreak of spruce budworms continues in a cyclical manner.